
5.3 Voice Disorders

5.3.1 Papillomatosis

◆ Key Features

- Papillomatosis is the most common benign neoplasm of the larynx.
- It is a human papillomavirus (HPV)-induced lesion.
- There is a low chance of malignant conversion.
- The mainstay of therapy is surgical resection, although adjuvant therapies may be helpful in severe cases.

Papillomatosis may affect any mucosal surface of the head and neck, but it has a predilection for junctions between ciliated respiratory and squamous mucosa. The most common sites are the nasal vestibule, oropharynx, nasopharyngeal surface of the soft palate, upper and lower limits of the laryngeal ventricle, and undersurface of the vocal folds. HPV is the etiologic agent, with some subtypes (HPV-6, HPV-11) predominating. Recurrent respiratory papillomatosis (RRP) typically presents with dysphonia, although some children, and even fewer adults, may present with airway compromise. Treatment is surgical resection, with cold steel dissection, microdebrider, or laser. Adjuvant chemotherapy agents may have some role in recalcitrant cases.

◆ Epidemiology

Laryngeal papilloma can be categorized into two subgroups: juvenile and adult onset. Juvenile usually occurs in children less than 5 years, with 25% presenting in infancy. Incidence of juvenile RRP is 4.3 per 100,000 in the United States. Children are frequently (75%) the firstborn, vaginally delivered offspring of teenage mothers. Males and females are equally affected. Transmission is maternal–neonatal and of higher risk in those with active HPV genital warts (condylomata), although other factors play a role. Adult-onset RRP presents at ages 20 to 40 years, with a 4:1 male/female ratio. Incidence is 1.8 per 100,000. In adults, the disease is likely to be sexually transmitted.

◆ Clinical

Signs and Symptoms

Children may present with airway obstructive symptoms, particularly if they are presenting very young. Children may also have a husky cry or dysphonia. Secondary to the rare nature of the disease and the pediatrician's inability to visualize the larynx, children are frequently misdiagnosed as having other airway problems such as asthma, bronchitis, or croup. Symptoms

are present on average 1 year before diagnosis. In some cases, children will present with emergent airway obstruction. Adults typically present with dysphonia, rarely with acute airway compromise. Symptoms are present on average for 6 months before diagnosis.

Differential Diagnosis

Hoarseness in children may be caused by vocal nodules, reflux disease, vocal fold immobility, laryngotracheobronchitis, laryngeal cysts, congenital laryngeal abnormality, or neurologic conditions. Upper airway compromise causes may include congenital laryngeal lesions, laryngeal cysts, vocal fold immobility, subglottic stenosis, a foreign body, and infectious processes, such as epiglottitis or laryngotracheobronchitis.

Hoarseness in adults may be caused by vocal fold nodules, reflux laryngitis, vocal fold cysts or polyps, leukoplakia, vocal fold neoplasms, sulcus vocalis, inflammatory laryngitis (e.g., tobacco abuse, steroid inhalers), vocal fold immobility, hypothyroidism, and systemic illnesses such as sarcoidosis or amyloidosis.

◆ Evaluation

Physical Exam

The physical exam should include a full head and neck examination. Attention should be made to the respiratory status of the patient, to assess whether acute interventions will be necessary to preserve the airway. Vocal quality should be noted. Indirect laryngoscopy can be done on adult patients. Flexible laryngoscopy may be done in adults and nondistressed children to assess location, extent, and functional limitations of the papilloma disease. Videostroboscopy can be useful, when available, to assess the impact of the papilloma on mucosal wave dynamics.

Imaging

Imaging has limited use, except in assessing for other issues causing airway compromise in children or assessing distal pulmonary papillomatosis. High-kilovoltage plain films or airway fluoroscopy may be helpful in this regard but do not specifically assist in the diagnosis of RRP.

Labs

No specific labs are helpful in RRP. Some otolaryngologists recommend HPV typing at the time of resection. This does not alter treatment but can offer some prognostic information, as HPV-11 patients tend to have more aggressive disease, more recurrences, more surgical procedures, and more use of adjuvant therapies.

Pathology

Papillomas contain a pedunculated, vascular, fibrous core with overlying nonkeratinized squamous epithelium. Multiple projections emanate off the central core, giving a frond- or wart-like configuration. Cellular atypia may occur in the epithelium and can be concerning for premalignant changes.

◆ Treatment Options

Medical

Although there are no primary medical therapies for RRP, adjuvant therapy to surgical resections may be necessary. Criteria for adjuvant methods include more than four procedures per year, rapid recurrence with airway compromise, and distant spread of disease. Approximately 10% of juvenile RRP patients require adjuvants.

The most commonly used therapy is recombinant α -interferon. This protein complex is a host defense to viral infection and immunomodulates the host into an antiviral condition. This has been shown to reduce the frequency of operative interventions. It is administered daily for 1 month, then tapered to three times weekly for at least 6 months. Side effects include flu-like symptoms, alopecia, leukopenia, coagulopathy, and neurologic side effects.

Indole-3-carbinol is an herbal supplement derived from cruciferous vegetables. The mechanism is unclear but is believed to be related to alterations in estrogen metabolism. Studies show a majority of patients receive partial to complete response. Dosages for children less than 25 kg are 100 to 200 mg daily and for adults 200 mg twice daily. Side effects include headache and dizziness.

Cidofovir is a cytosine nucleotide analog antiviral agent, designed for herpetic viruses and cytomegalovirus. Intralesional injections have shown good response in some patients. Concern for promoting progression to squamous cell carcinoma has been raised, but such progression is not proven.

Acyclovir has been used systemically, but the benefits are not well defined.

Photodynamic therapy utilizes the uptake of hematoporphyrins by papilloma to sensitize the tissue to red laser light. Disease progression is improved, but remission is not achieved.

Quadrivalent HPV vaccines have recently been released for HPV subtypes 6, 11, 16, and 18, with the indication for treating young girls before sexual activity to reduce the rate of cervical cancers. The hope exists that this use will influence the rate of laryngeal papilloma in future generations, and it may even be applied to males in the future.

Surgical

Surgical resection via microlaryngoscopy remains the mainstay of therapy. Recurrences are typical and multiple procedures are the norm. Juvenile RRP tends to be more aggressive and require more surgeries, likely related to increased growth rate of the RRP and the smaller dimensions of the juvenile larynx prompting earlier intervention for recurrence. Techniques for removal include several modalities and are influenced heavily by surgeon preference. Cold steel dissection of the papilloma may be useful for small isolated lesions, but not diffuse lesions. The CO₂ laser has been the workhorse of RRP for many years. Vaporization of the lesions is a viable treatment option. The laryngeal microdébrider can also be used; it is favored by some surgeons over laser, as it may have less "peripheral damage" given its lack of thermal injury and does not require special intraoperative laser precautions. Several new lasers have been utilized in

RRP, even in the office setting via flexible scopes. These include flexible CO₂ and pulsed dye lasers.

◆ Outcome and Follow-Up

Recommendations regarding postoperative care vary by surgeon (e.g., voice rest). Reflux medications are recommended by many to reduce postoperative scarring exacerbated by any acid exposure, as well as to reduce the potential cofactor of acid exposure in RRP regrowth.

By its very name, RRP is a recurrent problem for patients. The course is variable, with some patients experiencing lifelong recurrences and others manifesting spontaneous remissions. Juvenile-onset RRP does seem to have a higher rate of remission as the children enter adolescence.

Some patients are concerned about spread of RRP to family members or sexual partners. There are no well-documented cases of patient-to-patient transmission of laryngeal RRP. Some theoretical concern exists for caregivers, however, with reported viable virus in laser smoke plumes.

5.3.2 Vocal Fold Cysts, Nodules, and Polyps

◆ Key Features

- Nonneoplastic disorders of the larynx include nodules, cysts, and polyps.
- Dysphonia is a common presenting complaint.
- Large polyps may rarely present with airway obstruction.

Nonneoplastic changes affecting the vocal folds are common causes of chronic hoarseness. An office exam and videostroboscopy can generally lead to an accurate diagnosis. Treatment may involve voice therapy, microscopic voice surgery (microlaryngeal surgery), and behavioral modifications. With accurate diagnosis, appropriate management, and patient compliance, treatment should be highly effective.

◆ Epidemiology

Benign and reactive laryngeal lesions are common disorders; true incidence is difficult to determine.

◆ Clinical

Signs and Symptoms

Patients with various nonneoplastic vocal fold disorders generally complain of hoarseness of variable duration. A history of voice abuse or violent coughing is common. Vocal nodules are almost always seen in young women

or young children with a history of voice abuse. Vocal fold polyps are more frequent in men. Stridor is occasionally associated with large vocal polyps. Vocal fold cysts present also with substantial voice changes and usually affect adults. Cysts are thought to be congenital or acquired and seem to arise in the setting of chronic irritation and inflammation after hemorrhage. Reinke's edema, or polypoid chondritis, is a distinct entity from vocal polyps or nodules and usually presents with a pitch disturbance (e.g., females being mistaken for males on the telephone). It is caused by tobacco abuse.

Differential Diagnosis

The major consideration is the possibility of laryngeal squamous cell carcinoma, especially in the setting of tobacco abuse. History and physical examination alone may be insufficient to differentiate early-stage cancer from certain benign conditions. Differential diagnosis includes nodules, cysts, polyps, Reinke's edema, granuloma, papillomatosis, reflux laryngitis, other chronic laryngitis forms, glottic sulcus vocalis, and squamous cell carcinoma.

◆ Evaluation

History

A specific history of severe coughing, anticoagulant use, voice abuse, vocal fatigue, singing, tobacco abuse, or reflux symptoms should be ascertained. Prior intubations, laryngeal trauma, or laryngeal surgeries must not be overlooked. It is important to elicit timing of onset.

Physical Exam

A complete head and neck exam is performed. Office endoscopy and videostroboscopy, if the latter is available, often enable a diagnosis. Vocal nodules are seen as a thickened bilaterally symmetric lesion involving the midmembranous vocal fold contact site, generally found in a young female patient. Vocal fold cysts are usually solitary and visible as a bulge in the affected fold, but when small may be confused with other subepithelial lesions. Characteristic changes in vibration are appreciable on stroboscopic exam. Vocal fold polyps can be variable in size and may be single or multiple. Generally, polyps arise near the vibratory margin of the vocal fold. A feeding vessel or evidence of hemorrhage may be obvious. Large pedunculated polyps are occasionally seen arising from the upper or lower surface of the vocal fold. Reinke's edema appears as symmetric pale thickening of the folds, often with dehydration and thick mucus. Glottic sulcus vocalis may be suggested by stroboscopic findings revealing a segment with reduced vibration, but microlaryngoscopy is often required for definitive diagnosis.

Imaging

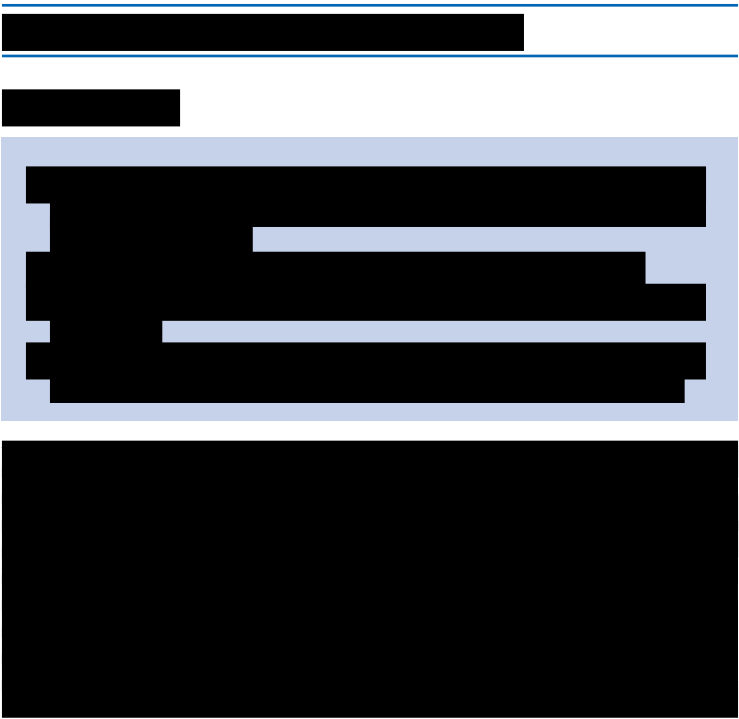
Radiologic examination is not indicated in most cases.

◆ **Treatment Options**

Vocal nodules usually respond to voice therapy alone. Vocal fold cysts may respond to voice therapy; if not, they are removed by microsurgery via cordotomy and microflap. Vocal fold polyps may respond to voice therapy if small but are usually treated with surgery followed by voice therapy and antireflux therapy. Surgery involves excision with microinstruments and the use of short-pulsed CO₂ laser coagulation of a feeding vessel, if present. Reinke's edema may be treated conservatively with voice rest and smoking cessation or treated surgically with submucosal evacuation via a cordotomy, generally placing the incision on the superolateral surface. In all cases, behavioral modification to address tobacco abuse, reflux, voice abuse, and vocal hygiene is important.

◆ **Outcome and Follow-Up**

In most cases, prognosis should be very good. Prevention of recurrence should involve follow-up monitoring of smoking and other behavioral factors discussed previously.



5.4.2 Dysphagia

◆ Key Features

- Dysphagia may reflect dysfunction at any level of the swallowing reflex from the oral cavity to the distal esophagus.
- Evaluation requires a detailed history, full head and neck examination, and often endoscopic evaluations and imaging studies.
- Swallowing therapy may be beneficial to those with upper aerodigestive tract dysfunction who have no medically or surgically correctable problem.

“Dysphagia” is a term encompassing a large area of clinical symptoms. Patients complain of “trouble swallowing.” A careful, detailed history may be necessary to elicit the exact nature of the patient's problems with swallowing. Dysfunction may be due to physical obstructive phenomena, neuromuscular weakness, or discoordination.

◆ Epidemiology

Dysphagia affects all age groups, dependent on the etiology of the symptom. The elderly are disproportionately affected, however. Some estimate that 50% of nursing home patients suffer with swallowing problems. The percentage of stroke patients who relate dysphagia is 50 to 75%.

◆ Clinical

Signs and Symptoms

Patients may relate very specific aspects of the swallowing reflex that are problematic for them, or they may simply have a generic complaint of trouble swallowing. Dysphagia is associated with symptoms including choking, gagging, globus, odynophagia, difficulty initiating swallow, drooling, aspiration, coughing, nasal reflux, and regurgitation. A history of pneumonia is concerning for chronic aspiration. Weight loss may be a sign of dysphagia significant enough to reduce caloric intake, or potentially a neoplastic process. Children with dysphagia may have similar symptoms but also prolonged feeding times, repeated swallow efforts, or unusual posturing during feeding.

Dysphagia may involve one or more of the phases of the swallowing reflex, including oral preparatory, pharyngeal, or esophageal phases. Specific symptoms may direct the examiner's attention to one of these sites.

Differential Diagnosis

- Neurologic disorder (e.g., myasthenia gravis, Parkinson's disease [PD], amyotrophic lateral sclerosis [ALS], multiple sclerosis, cerebral palsy,

mental retardation or developmental delay, muscular dystrophy, postpolio syndrome, cranial neuropathies)

- Neurologic injury (e.g., cerebrovascular accident, traumatic brain injury, neurosurgical or skull base procedures with cranial neuropathies)
- Obstruction (e.g., upper aerodigestive tract neoplasms, esophageal web, strictures, achalasia, foreign body, cricopharyngeal dysfunction, cervical spine osteophytes)
- Other (e.g., xerostomia, gastroesophageal reflux disease [GERD], presbyphagia or presbyesophagus, neck irradiation, cleft lip/palate, hypopharyngeal diverticulum, tracheotomy)

◆ Evaluation

Physical Exam

The physical exam should include all components of the upper aerodigestive tract. The oral cavity should be examined and note made of labial competence; salivary function; tongue mobility, symmetry, and strength; palatal motion; mucosal lesions or masses; and gag reflex. The pharyngeal examination should include mirror examination and/or flexible nasopharyngoscopy. In addition to examining for symmetry, mass effect, and mucosal lesion, the presence of pooled secretions is a key finding. Saliva and liquid or food residue in the vallecula, piriform recesses, and postcricoid area indicate either an obstruction of material passage somewhere between the cricopharyngeus and the stomach, or a sensory defect of the hypopharynx and larynx.

Laryngeal examination is a component of the pharyngeal evaluation. A “wet voice” may indicate retained secretions in the hypopharynx and laryngeal introitus. The sensation, general mobility, and ability to completely close the vocal folds should be assessed. Neck examination should note symmetry, contour, presence of masses, and presence of laryngeal elevation with swallow. Normal presence of laryngeal crepitus (easy mobility and click encountered on moving the larynx over the cervical spine) should be elicited. Cranial nerves (CNs) should be assessed, particularly those associated with the swallow mechanism: CNs V, VII, IX, X, and XII.

Imaging

Plain film radiography has a limited role in the assessment of dysphagia. Cervical films may demonstrate foreign bodies, cervical osteophytes, and air–fluid levels in the cervical esophagus or diverticulum. Chest radiographs may show pneumonia or evidence of chronic aspiration, as well as esophageal air–fluid levels. The mainstay of evaluation is the modified barium swallow (MBS), or rehabilitation swallow. This test is typically done in conjunction with a speech pathologist. Cinefluoroscopic examination involves having the patient swallow contrast containing a coated material of different consistencies. This allows for an anatomic evaluation of the upper aerodigestive tract but also a functional assessment of bolus preparation, transfer, and transit. Patients may vary in their ability to manage various

consistencies of food; the MBS enables identification of this fact, as well as information about therapeutic consistency changes. Retained material may be noted. Additionally, laryngeal penetration and aspiration may be identified. Rehabilitation maneuvers may be tried under fluoroscopy by the speech pathologist and feedback about their effectiveness can be immediately ascertained. Computed tomography (CT) and magnetic resonance imaging (MRI) may be necessary to investigate asymmetry for the pharynx and esophagus or when tumor is suspected.

Other Tests

Fiberoptic endoscopic evaluation of swallowing (with or without sensory testing), or FEES, is another technique available to assess the swallow mechanism. This can be an alternative to an MBS. A flexible endoscopy is passed through the nose and positioned to visualize the oropharynx and hypopharynx and larynx during swallowing trials. Trained substances are colored with food dye, and the examination is recorded to ease assessment of rapid events and subtle findings. Sensory testing is accomplished via an air pulse delivered through the flexible laryngoscope and identification of the presence and strength of the laryngeal adductor reflex. Formal esophagoscopy is useful if esophageal pathologies are suspected. This technique also allows for biopsy and sometime interventions. Some otolaryngologists have adopted in-office, nonsedated transnasal esophagoscopy for assessing the esophagus. Esophageal manometry and pH probe testing may be adjuncts to assess for esophageal dysmotility and reflux issues.

◆ Treatment Options

Medical

Dysphagia secondary to nonobstructive phenomena is typically managed with therapy techniques. Most of these techniques require voluntary action on the part of the patient, although some require little patient cognition. Techniques may involve head and neck positioning during the swallow, food consistency changes, and sensory enhancement therapy. A speech pathologist with an interest in swallow rehabilitation should be consulted. Many patients perform better with various consistencies of food: dietary modification and thickening agents may help a patient maintain oral intake. Nutritional supplements may be useful in the nutritionally compromised patient. If reflux is a component of the dysphagia or is thought to be the underlying etiology, medical therapy may be tried.

Surgical

Surgery may be directed to obstructive phenomena such as tumors, strictures, webs, cricopharyngeal hypertonicity, or hypopharyngeal diverticulum. Vocal fold immobility that contributes to aspiration issues may benefit from medialization thyroplasty. Procedures for chronic aspiration exist (see **Chapter 5.4.3**). If oral intake is deemed unsafe due to aspiration, feeding gastrostomy may be necessary.

◆ Outcome and Follow-Up

The etiology largely determines the treatment outcome and follow-up.

5.4.3 Aspiration

◆ Key Features

- Aspiration involves the passage of secretions or material into the lower airways.
- Aspiration is an important source of morbidity in the neuromuscularly impaired and debilitated.
- The most common cause of aspiration is neuromuscular dysfunction.
- Chronic aspiration may have severe medical consequences, and medical, sometimes surgical, interventions need to be enacted.

In general, the lower airway should not be exposed to upper aerodigestive tract secretions or materials. Aspiration of small amounts of material during sleep (reported in 50% normal patients) may be tolerated if the tracheobronchopulmonary system clearing mechanisms are functional. Large episodes of aspiration or chronic aspiration may yield complications whose severity is determined by the nature and volume of the material aspirated. Aspiration may be primary (swallowed dietary materials or secretions) or secondary (regurgitated diverticulum or gastric contents).

◆ Clinical

Signs and Symptoms

The presence of chronic aspiration may be evident to the patient and health-care providers or may be “silent,” with no cough generated. Dysphagia is intimately associated with aspiration, as most patients with noted aspiration will also complain of “trouble swallowing” (see **Chapter 5.4.2**). Symptoms of chronic aspiration include choking or coughing when swallowing (may be with specific types of dietary intake, all food, or even the patient’s own secretions). Some patients may also complain of chronic cough independent of swallowing, related to the bronchopulmonary complications of the chronic aspiration. Productive cough, fevers, and dyspnea in a patient with dysphagia is concerning for this issue. Also, recurrent lower respiratory infections in a patient with predisposing comorbidities should make the clinician suspicious of this issue.

Differential Diagnosis

Some conditions may share respiratory symptoms with the complications of chronic aspiration. The key differential diagnosis is for those diseases that may predispose to chronic aspiration:

- Neuromuscular
 - Degenerative diseases
 - Alzheimer's disease
 - Amyotrophic lateral sclerosis (ALS)
 - Multiple sclerosis
 - Progressive supranuclear palsy
 - Parkinson's disease (PD)
 - Huntington's disease
 - Disorders
 - Myasthenia gravis
 - Myopathy (dermatomyositis, inclusion body myositis)
 - Muscular dystrophy
 - Poliomyelitis and postpolio syndrome
 - Guillain-Barré's syndrome
 - Cranial neuropathies
- Central nervous system (CNS) injury
 - Cerebrovascular accident
 - Head trauma
 - Anoxic brain injury
 - Postsurgical
 - Encephalitis/meningitis
 - Cerebral palsy
- Upper aerodigestive tract disorders
 - Vocal fold immobility
 - Stricture (web, reflux, caustic ingestion, postsurgical)
 - Zenker's diverticulum
 - Cricopharyngeal dysfunction
 - Postirradiation/chemoradiation
 - Postsurgical dysfunction
 - Neoplasms
 - Achalasia
 - Reflux
- Other
 - CNS/skull base neoplasm
 - Severe deconditioning or systemic illness
 - Intoxication
 - Mental retardation or developmental delay
 - Presence of a tracheotomy tube

◆ Evaluation

After a detailed history and physical examination (including nasopharyngoscopy), several options exist for evaluation. A bedside swallow evaluation by a speech pathologist is reasonable for those with mild dysphagia, but secondary to the morbidity that may accompany aspiration; if aspiration is suspected, an objective evaluation is warranted.

Imaging

Modified barium swallow (MBS) has been the traditional imaging study for evaluation of dysphagia and aspiration. This test may demonstrate the passage of contrast material into the trachea and bronchi. Laryngeal penetration or retained contrast within the piriform recesses or vallecula are concerning because of their aspiration potential. Therapeutic maneuvers may be performed under cinefluoroscopy and their impact determined immediately, thus aiding in care planning.

Scintigraphy has been proposed to demonstrate and quantify aspiration. Overall, it offers few advantages beyond MBS and fiberoptic endoscopic evaluation of swallowing (FEES) (see Other Tests). Chest radiography (plain film or computed tomographic [CT] radiography) may be helpful in demonstrating pneumonia and chronic pulmonary disease stemming from aspiration (e.g., bronchiectasis).

Other Tests

FEES involves the transnasal endoscopic monitoring of dyed food and liquid ingestion. Observations are made regarding premature leakage, retained materials, laryngeal penetration, or aspiration. This study is typically video-recorded so that slow motion review may be performed to assess for subtle evidence of swallowing dysfunction.

◆ Treatment Options

Initial management in a patient who has concerns for chronic aspiration is to ensure NPO (nothing by mouth) status, to minimize the volume of potential aspirate until confirmation can be made. Good oral care and speech therapy assessment are appropriate in all patients. Alternative nutrition routes such as nasogastric tubes or percutaneous feeding conduits should be considered, depending on clinical circumstances, for those with prolonged inadequate nutrition.

Medical

A variety of speech therapy techniques are available to assist in safe swallowing. These include consistency changes (e.g., thickening agents, alternating consistencies), as well as positioning maneuvers (e.g., chin tuck, head turn) and swallowing maneuvers (e.g., double swallow, swallow–cough, swallow–cough–swallow, effortful swallow). If a patient is deemed to be inappropriate for any oral intake, alternative feeding conduits should be utilized. If aspiration of secretions is an issue, surgical procedures may be necessary.

Table 5.4 Surgical management of chronic aspiration

Procedure	Description	Advantages	Disadvantages
Irreversible			
Narrow-field laryngectomy	Removal of larynx, preserving hyoid and strap muscles	Definitive separation of respiratory and digestive tracts	Definitively irreversible Negative psychosocial features for patients Transcervical approach Loss of phonation
Glottic closure	Median thyrotomy with suture closure of true and false vocal folds	High success rate Theoretical, but difficult reversal	Transcervical approach Tracheotomy Loss of phonation
Subperichondrial cricoidectomy	Subperichondrial resection of cricoid with closure subglottic mucosa and strap muscle reinforcement	High success rate Possible under local anesthesia	Transcervical approach Tracheotomy Loss of phonation Fistula potential
Reversible			
Laryngotracheal separation	Transection of the trachea with creation of tracheostoma and closure of subglottis	Among the most dependable of reversible procedures	Transcervical approach Tracheotomy Loss of phonation Fistula potential
Laryngotracheal diversion	Transection of the trachea with creation of tracheostoma and anastomosis of proximal trachea to esophagus	Among the most dependable of reversible procedures “Aspirated” secretions into proximal trachea diverted to esophagus	Transcervical approach Tracheotomy Loss of phonation Fistula potential
Epiglottic flap laryngeal closure	Larynx is closed by denuding edges of epiglottis and aryepiglottic folds and suturing epiglottis over the glottis	Potential speech preservation if posterior glottis is kept open	Transcervical approach Tracheotomy Possible dehiscence 50% prevention of aspiration Supraglottic stenosis after reversal

continued

Table 5.4 Surgical management of chronic aspiration (*Continued*)

Double-barrel tracheotomy	Transection of trachea with externalization of both ends of the trachea	Decreased fistulization	Transcervical approach Skin contamination with secretions
Partial cricoidectomy	Submucosal resection of posterior cricoid with cricopharyngeal myotomy and tracheotomy, serves to narrow laryngeal inlet and widen esophageal inlet	Preserves the voice	Transcervical approach Tracheotomy Persistent aspiration
Vertical laryngoplasty	Denuding of lower edge of the epiglottis, aryepiglottic folds, arytenoids with vertical tubing of the epiglottis, leaving superior opening	Preservation of voice Described for total glossectomy patients to reduce need for laryngectomy	Transcervical approach Persistent aspiration
Endolaryngeal stenting	Endolaryngeal stent placed to occlude the larynx	Simple to place	Leakage around the stent and aspiration Endolaryngeal injury Airway compromise if tracheotomy tube displaced Patient discomfort

Medical therapy should also involve treatment for the complications of the aspiration. These may include intubation, ventilator support, bronchoscopy, antibiotics, and pulmonary toilet.

Surgical

If medical management fails to correct the aspiration problem or is deemed inadequate, several surgical techniques are available. Placement of a tracheotomy tube facilitates pulmonary toilet and, with a cuffed tube inflated, may decrease aspiration events. An inflated cuffed tracheotomy tube, however, will not prevent aspiration. The presence of a tracheotomy tube may worsen or even cause aspiration: the laryngeal elevation accompanying a swallow is limited by tracheal tethering by the tube, and laryngeal sensation gradually decreases when airflow through the glottis ceases. A speaking valve may aid in swallowing by improving subglottic pressures necessary to clear secretions, and maintenance of laryngeal sensation. Directed surgical procedures may be helpful in reducing aspiration (e.g., chemical or surgical cricopharyngeal myotomy for cricopharyngeal dysfunction, vocal fold medialization for vocal fold immobility).

Laryngeal surgical management of chronic aspiration may be classified as reversible and irreversible (**Table 5.4**). Each of these techniques has advantages and disadvantages that may influence its use. Reversible procedures all have the potential advantage of being reversible if the underlying aspiration condition improves sufficiently.

◆ Complications

Complications may include chronic lung disease and pulmonary fibrosis.

◆ Outcome and Follow-Up

Outcome depends on the etiology of aspiration.

5.5 Acid Reflux Disorders

◆ Key Features

- Laryngopharyngeal reflux is distinct from classic gastroesophageal reflux disease (GERD).
- The work-up may involve an endoscopy, barium studies, and a pH probe.
- Extraesophageal manifestations relate to the pharynx, larynx, and lungs.

◆ Epidemiology

Twenty-five to forty percent of healthy adult Americans experience symptomatic gastroesophageal reflux disease (GERD) manifested as heartburn. Laryngopharyngeal reflux (LPR) typically does not cause heartburn or esophagitis. Certain foods, medications, hormones, and physical states (e.g., obesity) can decrease the pressure of the lower esophageal sphincter, thus worsening LPR or GERD symptoms.

◆ Clinical

Signs

GERD: esophagitis, Barrett's esophagus

LPR: coughing, throat clearing, laryngeal erythema

Symptoms

GERD symptoms typically include postprandial heartburn and regurgitation. LPR symptoms may be more subtle and include hoarseness, globus sensation, dysphagia, chronic throat clearing, halitosis, sensation of postnasal drainage, chronic cough, laryngospasm, and otalgia.

◆ Evaluation

A suggestive history and exam findings usually prompt a “therapeutic trial” of behavioral changes and twice-daily use of a proton pump inhibitor. Ancillary testing should be employed if this regimen fails or if symptoms are atypical or worsening, due to risk of malignancy.

Physical Exam

Nasolaryngopharyngoscopy may reveal laryngeal erythema and edema of arytenoids and interarytenoid space, laryngeal pachydermia, and vocal fold granulomas.

Imaging

A barium swallow study is useful for initial screening and to identify structural abnormalities.

Labs

Laboratory tests are seldom useful.

Other Tests

A dual-channel 24-hour pH monitoring probe is the gold standard. Esophagogastroduodenoscopy identifies the presence and severity of esophagitis and the possible presence of Barrett's esophagus. The role of screening endoscopy is evolving. With the advent of the transnasal esophagoscope (TNE), the routine screening of reflux patients for occult esophagitis or Barrett's esophagus has become more commonplace.

Pathology

Transient lower esophageal sphincter relaxation is a manometric finding. The Reflux Symptom Index (RSI), a patient-completed survey, is useful for scoring or grading.

◆ Treatment Options

Medical

Behavioral modification: weight loss, smoking cessation, avoid eating before sleep, avoid caffeine, alcohol, peppermint, chocolate, and spicy and acid foods.

H₂ receptor antagonists: ranitidine (e.g., Zantac, Boehringer Ingelheim Pharmaceuticals, Ridgefield, CT) 150 mg orally twice daily. These agents block histamine at the H₂ receptors, particularly those in the gastric parietal cells, to inhibit acid secretion. Tachyphylaxis may develop to H₂ blockers after several weeks of use.

Proton pump inhibitors (PPIs): esomeprazole (e.g., Nexium, Pfizer, New York, NY) 20 to 40 mg orally twice daily. These agents inhibit gastric acid secretion by inhibition of the H⁺/K⁺-ATPase system in the gastric parietal cells.

Prokinetics: metoclopramide (Reglan, Ani Pharmaceuticals, Baudette, MN) 10 mg orally four times daily. These agents increase lower esophageal sphincter pressure to help reduce reflux and also accelerate gastric emptying.

Surgical

About 20% of patients have a progressive form of reflux disease and may develop severe complications. For these patients, surgical treatment should be considered. The indications for this include Barrett's esophagus; strictures; respiratory manifestations (e.g., cough, wheezing, aspiration); ear, nose, and throat manifestations (e.g., hoarseness, sore throat, otitis media); and dental manifestations (e.g., enamel erosion).

With laparoscopic fundoplication, symptoms resolve in ~92% of patients.

◆ Complications

Complications of untreated or resistant GERD include esophagitis, pneumonia, asthma, and interstitial lung fibrosis. Barrett's esophagus is one of the most serious complications of GERD, because it may progress to cancer.

◆ Outcome and Follow-Up

Patients should understand that there is a need for long-term maintenance therapy. Many physicians advocate mandatory esophagoscopy, if patients continue to experience symptoms, to rule out Barrett's esophagus changes and cancer. Others advocate screen endoscopy in patients with a positive response to PPI treatment.